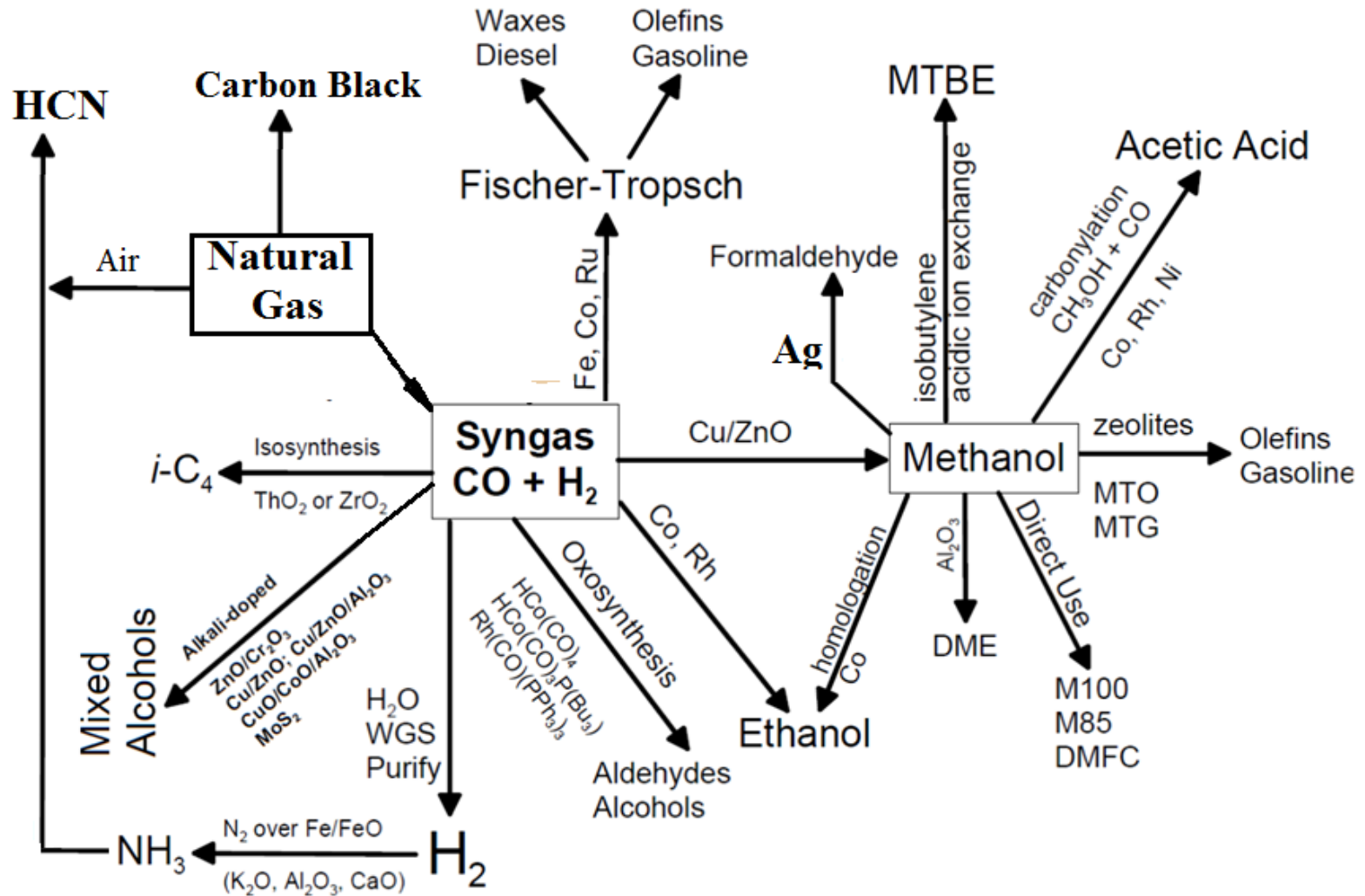


# Petrochemical Technology

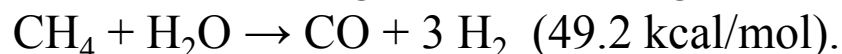
## **Chemicals from Natural Gas**

# Chemicals from Natural Gas



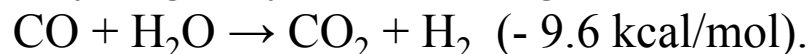
# Hydrogen Production from Natural Gas

**1: Reforming:** This involves catalytic reaction of methane with steam at temperatures in the range of 760 to 820°C according to the following reaction:



The reaction is endothermic and is carried out by passing the gas through catalyst filled tubes in a furnace. The catalyst is usually in the form of hollow cylindrical rings ranging upto  $\frac{3}{4}$  in. in diameter. It consists of **25-40% Nickel oxide on low-silica refractory base.**

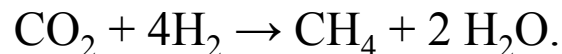
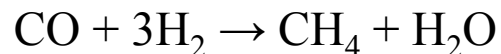
**2. Shift Conversion:** More steam is added to convert the CO from step 1 to an equivalent amount of hydrogen by the following reaction:



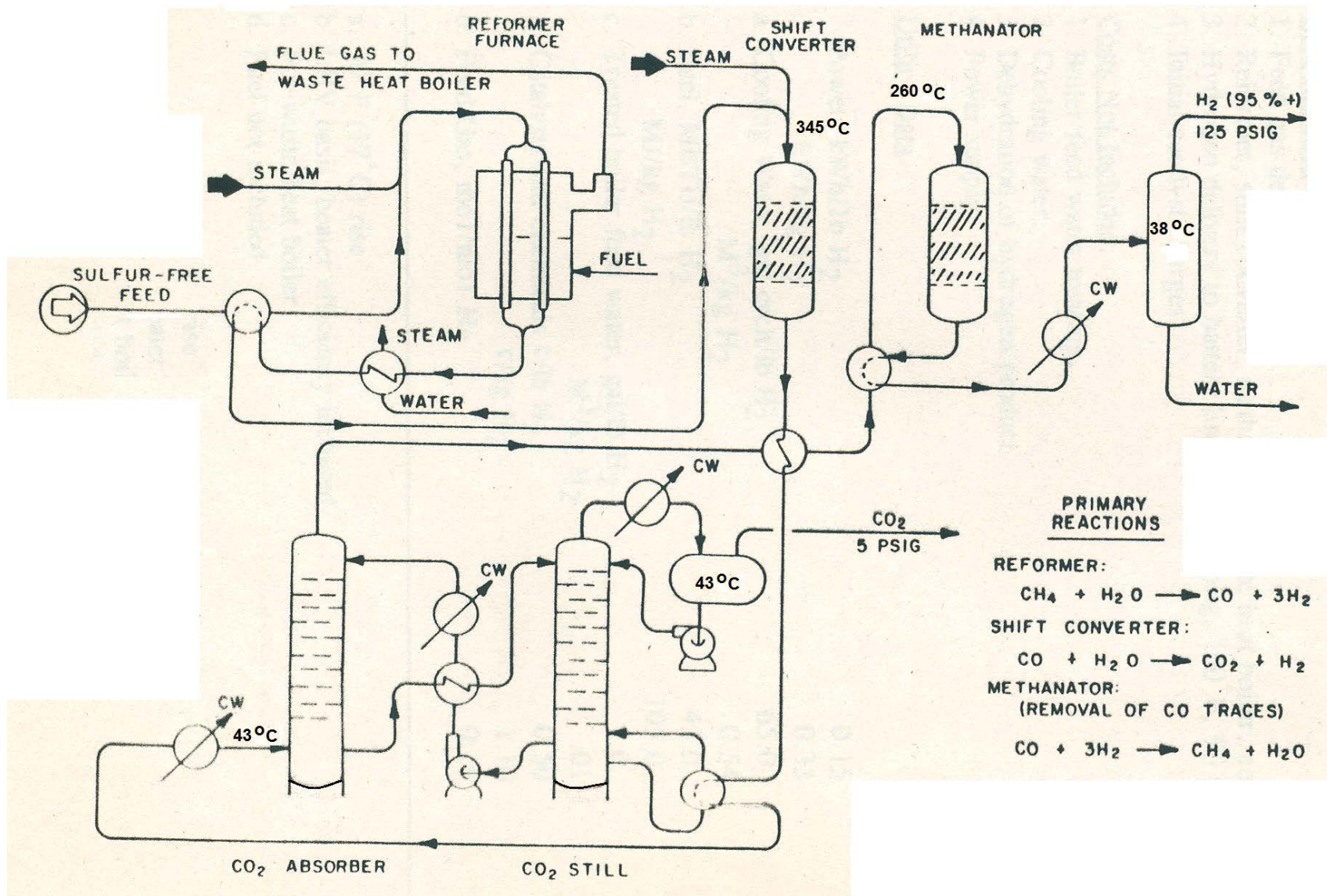
This is an exothermic reaction and is conducted in a fixed-bed catalytic reactor at about 345°C. Multiple catalyst bed in one reactor with external cooling between beds are commonly employed to prevent the temperature from getting too high as this would adversely affect the equilibrium conversion. **The catalyst used is a mixture of chromium and iron oxide.**

**3. Gas Purification:** The third step is the removal of CO<sub>2</sub> by absorption in a circulating amine or hot potassium carbonate solution. Several other treating solutions are in use. The treating solution contacts the hydrogen and CO<sub>2</sub> gas in an absorber containing about 24 trays or the equivalent amount of packing. Carbon dioxide is absorbed in the solution which is then sent to still for regeneration.

**4. Methanation:** In this step, the remaining small quantities of CO<sub>2</sub> and CO are converted to methane by the following reactions:



This step is also conducted in a fixed bed catalytic reactor at temperatures of about 425°C. Both reactions are exothermic and, if the feed concentrations of CO and CO<sub>2</sub> are more than 3%, it is necessary to recycle some of the cool exit gas to dissipate the heat of reaction. The **catalyst contains 10-20% nickel on a refractory base.**



**Flowsheet for Production of Hydrogen from Natural Gas**

# Synthesis Gas (Syngas) Production from Natural Gas (NG) by Autothermal Reforming (ATR)

## Reactions:

Reforming reaction:

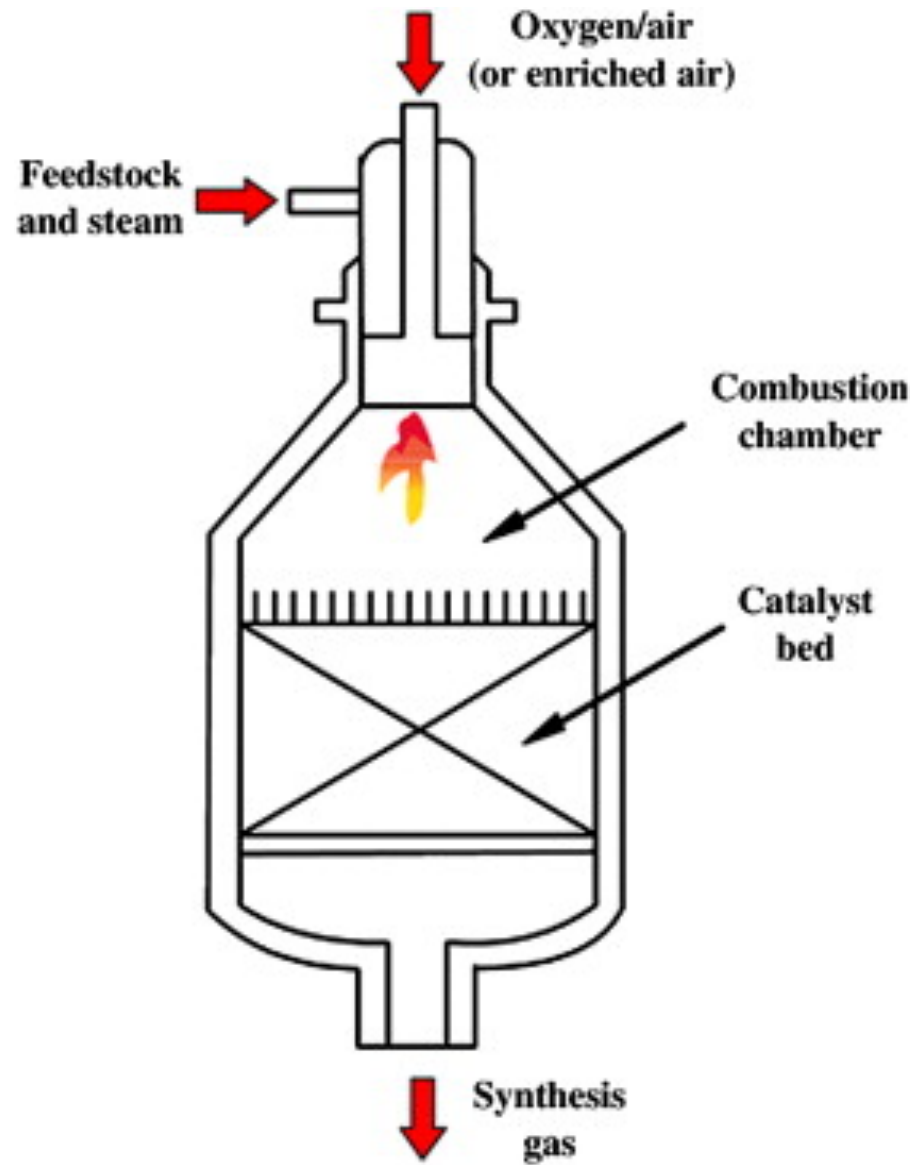


Combustion reactions:

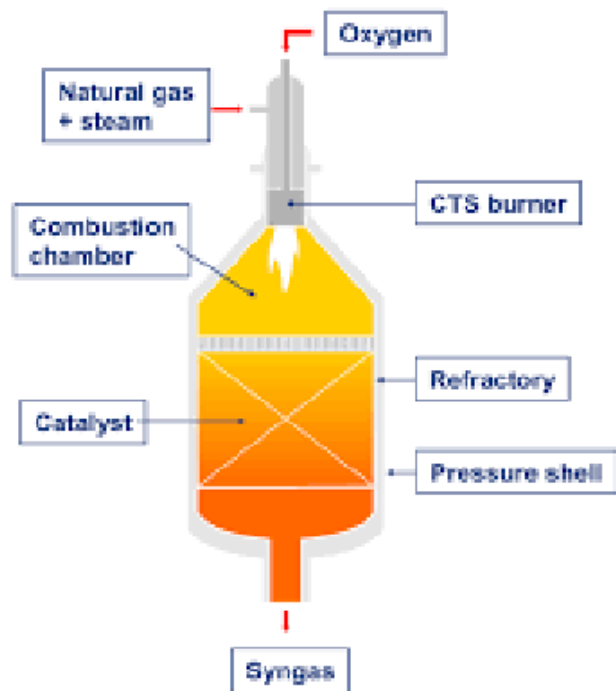


**Reaction Conditions:** Temperature, 760-820°C; Pressure, 30-35 atm

**Catalyst:** Nickel oxide (25-40%) on low silica refractory base



## Autothermal Reformer

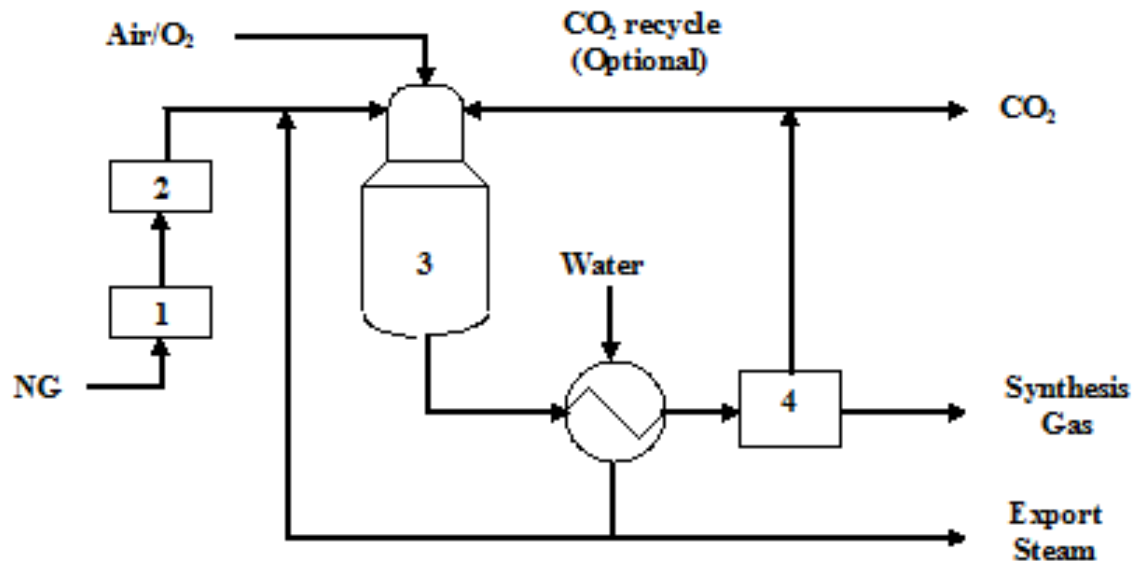


The autothermal reforming reactor consists of three zones: (1) the burner, where the feed streams are mixed in a turbulent diffusion flame; (2) the combustion zone, where partial oxidation reactions produce a mixture of carbon monoxide and hydrogen; and (3) the catalytic zone, where the gases leaving the combustion zone attain thermodynamic equilibrium. Key elements in the reactor are the burner and the catalyst bed – the burner provides mixing of the feed streams and the natural gas is converted in a turbulent diffusion flame:



When carbon dioxide is present in the feed, the  $\text{H}_2/\text{CO}$  ratio produced is on the order of 1:1, but when the process employs steam, the  $\text{H}_2/\text{CO}$  ratio produced is 2.5:1.

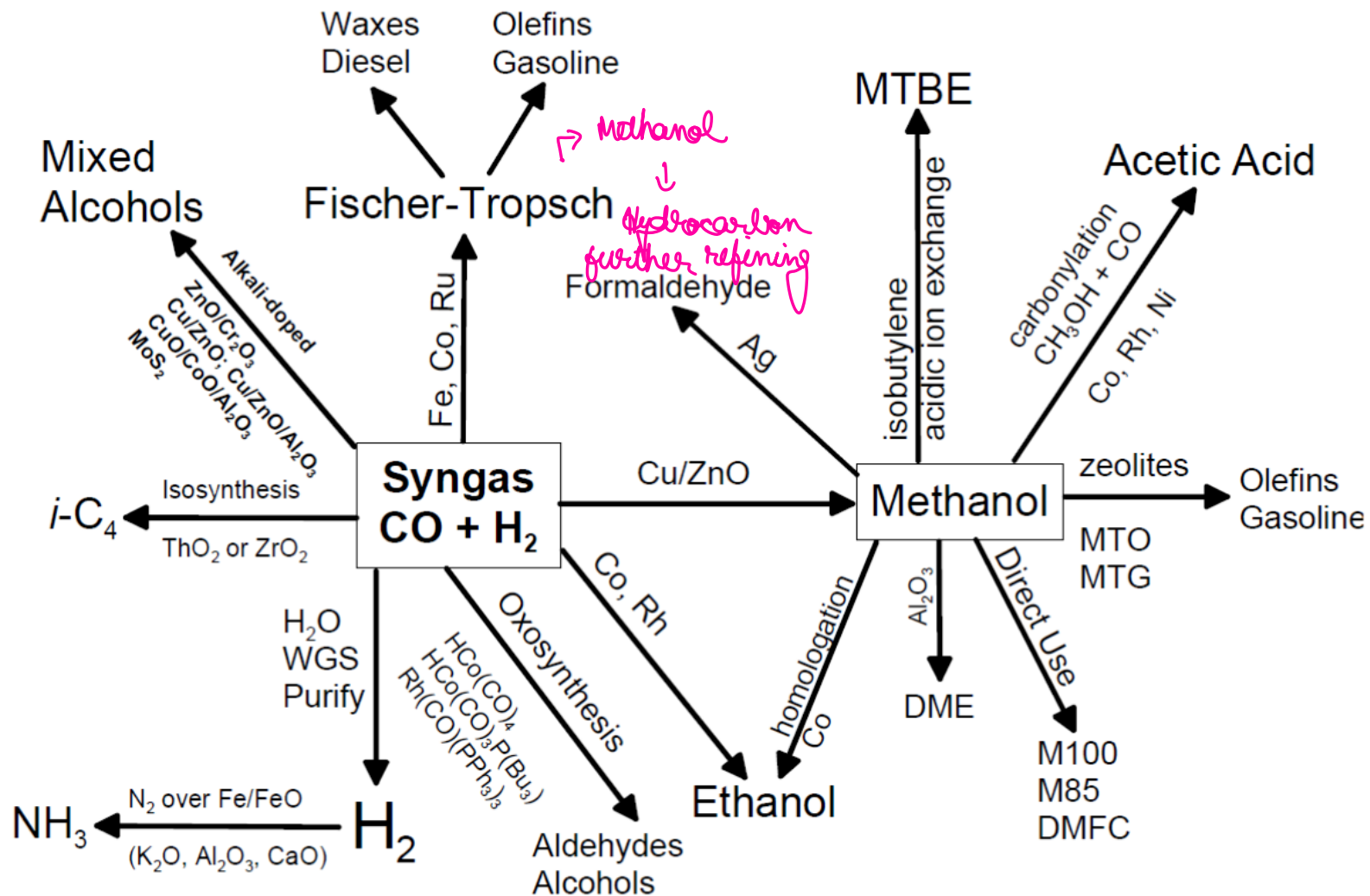




**1: Preheater; 2: Desulfurizer; 3: Autothermal Reformer; 4: CO<sub>2</sub> Removal unit**

Carbon dioxide can be recycled to adjust the H<sub>2</sub>/CO ratio in the product gas. With natural gas as the feedstocks, the H<sub>2</sub>/CO ratio in the product gas ranges from about 2.7 (for no CO<sub>2</sub> recycle) to 1.6 (for full CO<sub>2</sub> recycle).

## **Flow Diagram for Synthesis Gas Production from Natural Gas by Autothermal Reforming**



## Examples of uses of syngas and methanol

# Chemicals from Synthesis Gas

*Slurry bed → mechanical agitator*

| Product                                   | Feedstock     | Research reactors                                          | Technology/Process conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------|---------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methanol</b>                           | Synthesis gas | Fixed bed, tubular reactor, fluidized bed & slurry bed     | <u>ICI technology</u><br>Fixed bed reactor with quenching<br>Temperature: 240°C<br>Pressure: 50 bar<br>$H_2/CO = 2.0$<br>Catalyst: Oxides of Zn, Cu & chromium oxide as promoter<br><i>↗ main active catalyst</i><br><i>↘ promoter</i>                                                                                                                                                                                                                                                                                                                        |
| <b>Dimethyl ether</b>                     | Synthesis gas | Fixed bed / <u>slurry bed</u> / liquid phase bubble column | <u>Haldor-Topsoe, Denmark</u><br>$H_2$ to CO ratio in syngas: 2.0<br>Catalyst: $\gamma$ -alumina and $\gamma$ -alumina modified with phosphates or titanates<br>Temperature: 290-400°C; Pressure: 80-120 bar<br>$H_2/CO = 2.0$<br>Feed : Natural gas<br><br><u>NKK Corporation, Japan</u><br>Slurry bed reactor<br>Temperature: 250-320°C; Pressure: 30-50 bar<br>$H_2/CO = 1.0$<br>Feed : Coal<br><br><u>Air Products &amp; Chemicals, USA</u><br>Bubble column<br>Temperature: ~ 250°C; Pressure: 50 bar<br>$H_2/CO = 2.0$<br>Feed : Coal, Natural gas etc. |
| F-T products (Diesel, gasoline, wax etc.) | Synthesis gas | Fixed bed, fluidized bed & slurry bed                      | <u>Sasol (South Africa)</u><br>Fluidized bed reactor, 320-360°C and 22 bar<br>$H_2$ to CO ratio in syngas: 2.0<br>Catalyst: Iron, cobalt, nickel and ruthenium metal based                                                                                                                                                                                                                                                                                                                                                                                    |

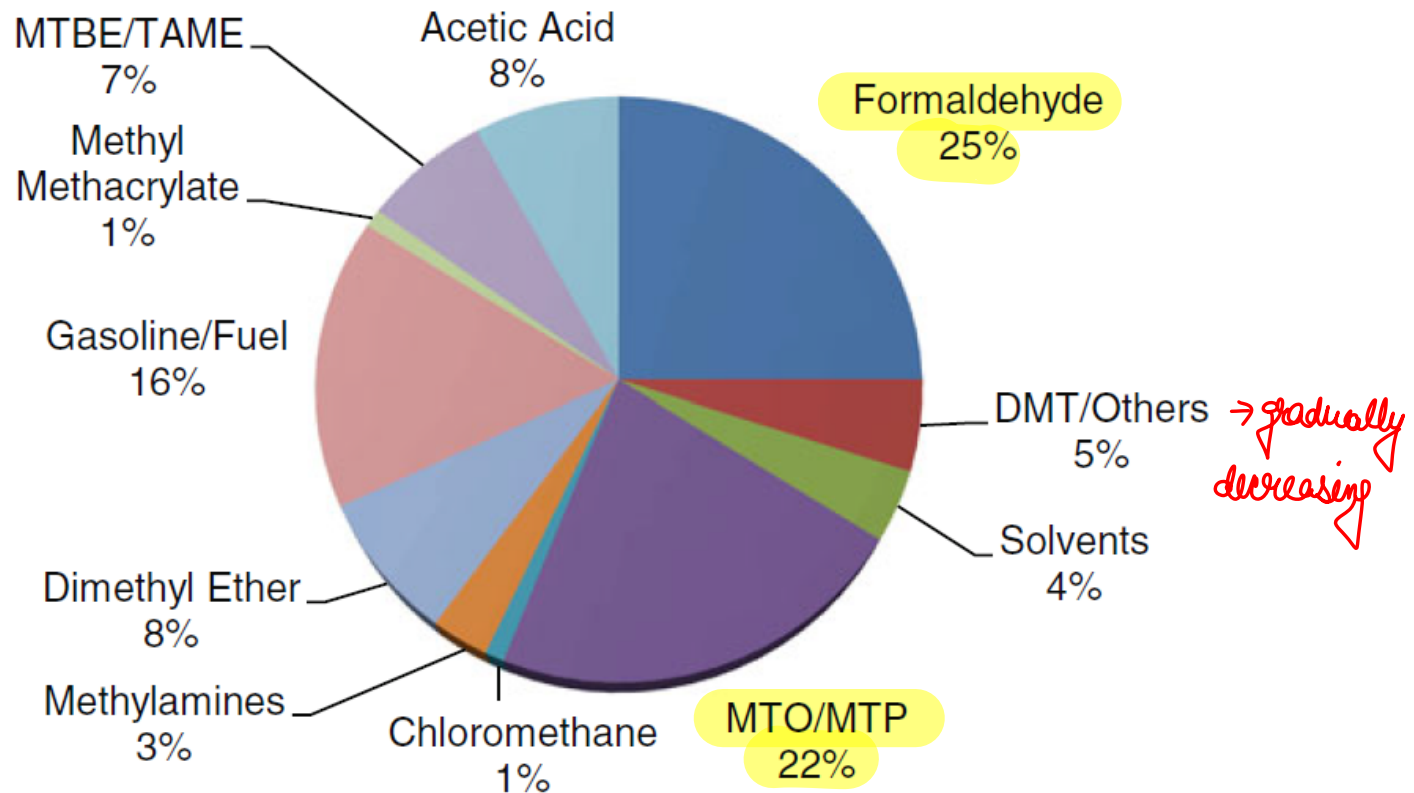
*Bubble reactor → high gas velocity is required, Don't use  $H_2$ , costly gas & more cost to give high velocity*

## Chemicals from Methanol

| Product                                                               | Feedstock | Research reactors                         | Technology/Process conditions                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------|-----------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthetic Gasoline                                                    | Methanol  | Fixed bed, fluidized bed                  | Mobil technology (MTG plant, New Zealand)<br>Fluidized bed reactor, 360-415°C and 22 bar<br>Catalyst: ZSM-5                                                                                                                                                                                                   |
| Olefins                                                               | Methanol  | Fixed bed, fluidized bed<br><br>Fluidized | Mobil technology (MTO plant, New Zealand)<br>Reactor: Fluidized bed reactor<br>Catalyst: ZSM-5<br>Temperature: 470-515 °C<br>Pressure: 3.0-4.5 bar<br>UOP Process<br>SAPO-34 catalyst → silica alumina phosphate                                                                                              |
| HCHO                                                                  | Methanol  | Fixed bed                                 | Ag- or Cu-gauge<br>Temperature: 450-600°C; Pressure: 1 atm.<br>MeOH/Air ratio: 0.3-0.5                                                                                                                                                                                                                        |
| CH <sub>3</sub> COOH<br><i>Monsanto process</i><br><i>Rh + Iodine</i> | Methanol  | CSTR                                      | Monsanto Process<br><u>Rhodium catalyst promoted with iodine</u> ; 180°C, 30-60bar<br>Cativa Process<br>Very similar to the Monsanto Process<br><u>Iridium-based catalyst</u><br>Advantages are <u>higher stability</u> of the <u>catalyst</u> <u>lower reaction rates</u> , <u>less liquid by-products</u> . |

# Syngas to Methanol

# Global Production: 111 MMTPA



## Methanol Demands and End Uses

DMT: dimethyl terephthalate, MTBE/TAME: methyl tert-butyl ether/tert-amyl methyl ether, MTO: methanol-to-olefins, MTP: methanol-to-propylene

# Syngas to Methanol

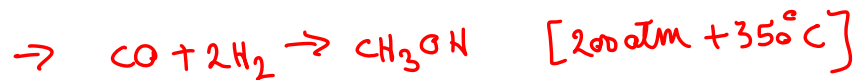
The transformation of synthesis gas to methyl alcohol is a process of major industrial importance. From the point of view of hydrocarbon chemistry, the significance of the process is the subsequent conversion of methanol to hydrocarbon (thus allowing F-T chemistry to become more selective).

*makes process more selective*

## Reaction:



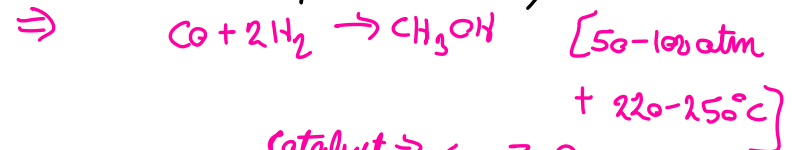
BASF 1923



*Catalyst  $\rightarrow \text{ZnO} + \text{chromia}$*

*$\rightarrow$  low specific activity so high temperature*

<sup>1960s</sup>  
**ICI** (low pressure version)



*Catalyst  $\rightarrow \text{Cu} + \text{ZnO}$*

An industrial process to produce methanol from carbon monoxide and hydrogen was developed by **BASF in 1923** using a zinc oxide – chromia catalyst. Since this catalyst exhibited low specific activity, high temperature was required. The low methanol concentration at this high temperature was compensated by using high pressures. This so-called high-pressure process was operated typically at 200 atm and 350°C.

The high-pressure process was superseded in the **1960s by the ICI** low-pressure process after introducing Cu-ZnO-based catalysts. Conditions applied in this process are 50-100 atm and 220-250°C.

A similar process was developed by **Lurgi** employing a CuO-ZnO catalyst. Since Cu-based catalysts are sensitive to sulfur and chlorine in the feed, a precondition for their industrial application is the use of high-purity synthesis gas prepared from methane.

\* 2 Imp 1 x

The Cu-ZnO binary catalyst has been known to exhibit high activity in methanol formation at low temperature, but fast deactivation prevented commercialization. The addition of a third component, however, resulted in stable catalyst activity. The third component is either  $\text{Al}_2\text{O}_3$  or  $\text{Cr}_2\text{O}_3$ . The primary functions of the non-copper components are to enhance dispersion of copper, and to maintain the active phase in a stable dispersion and with suitable physical strength. Cu-ZnO- $\text{Al}_2\text{O}_3$  and Cu-ZnO- $\text{Cr}_2\text{O}_3$  are usually the most important industrial catalysts, prepared by co-precipitation.

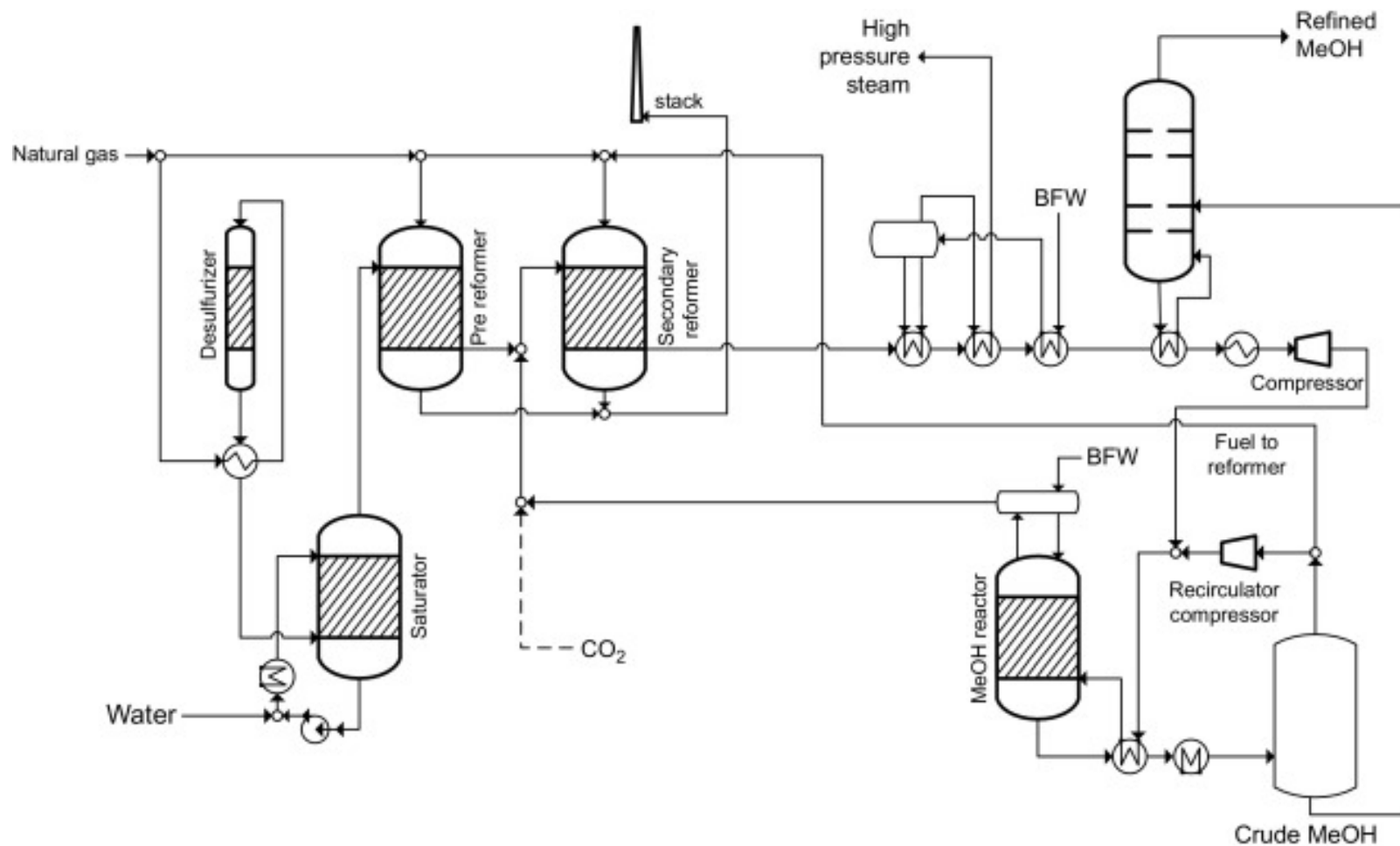
A significant advance was made when a two-stage precipitation was suggested in which  $\text{ZnAl}_2\text{O}_3$ , a crystalline zinc-aluminate spinel was prepared prior to the main precipitation of Cu-Zn species. This alteration resulted in an increase in catalyst stability for long-term performance with respect to deactivation. Catalyst lifetimes industrially are typically two years.

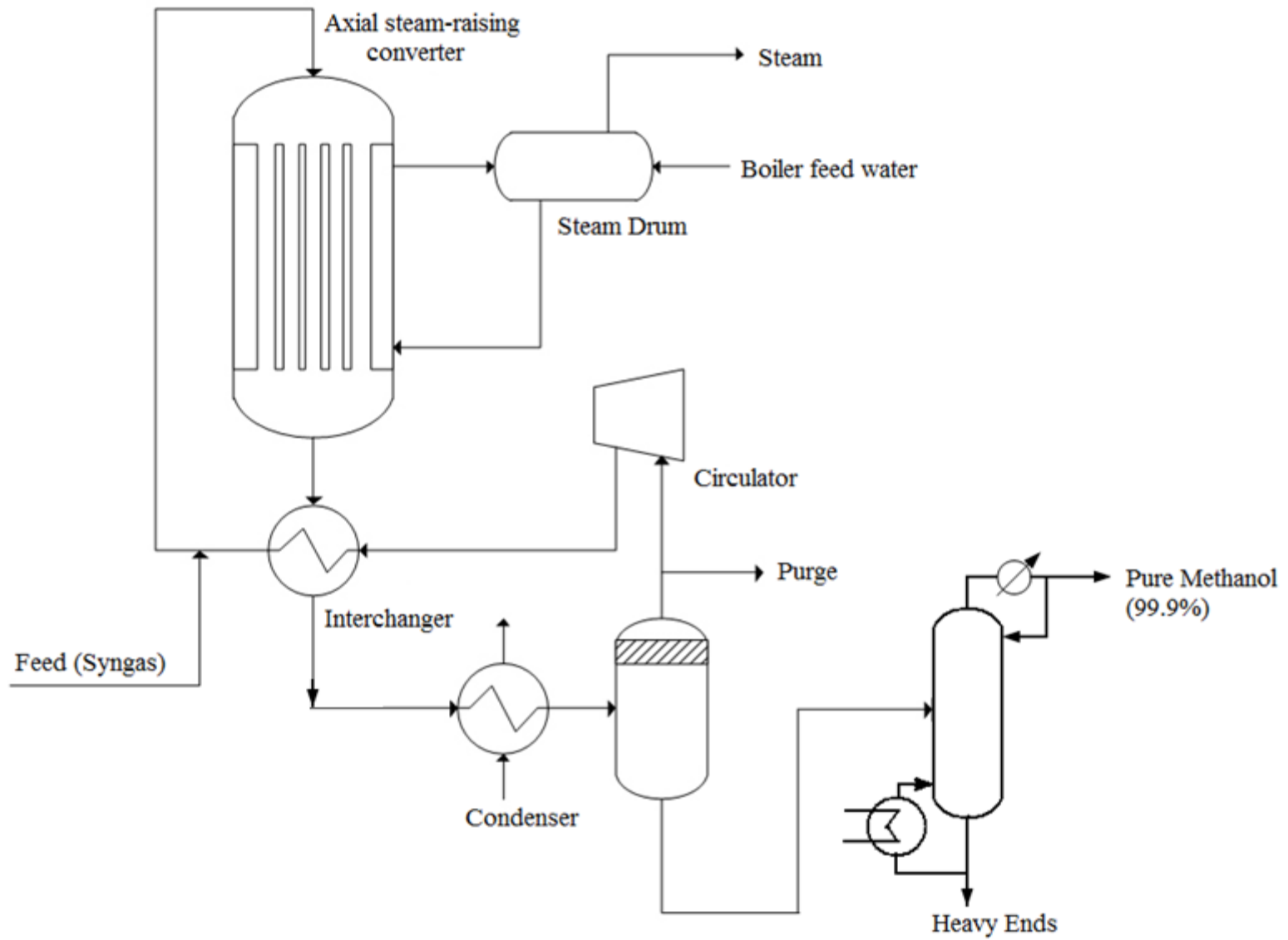
At present, the low-pressure technology dominates the worldwide production of methanol.

The low-pressure copper catalysts are very selective for the synthesis of methanol. Under industrial conditions on  $\text{Cu-ZnO-Al}_2\text{O}_3$  catalyst, the selectivity is typically greater than 99%. The impurities formed include hydrocarbons, higher alcohols, ethers, ketones and esters. These as well as any water formed can easily be removed by distillation to give very pure methanol.

## Metal composition of catalysts provided by different manufacturers

| Manufacturer            | Cu wt% | Zn wt% | Al wt% | Other (wt%)          |
|-------------------------|--------|--------|--------|----------------------|
| IFP                     | 45-70  | 15-35  | 4-20   | Zr (2-18)            |
| ICI                     | 20-35  | 15-50  | 4-20   | Mg                   |
| BASF                    | 38.5   | 48.8   | 12.9   |                      |
| Shell                   | 71     | 24     |        | Rare Earth oxide (5) |
| Sud Chemie              | 65     | 22     | 12     |                      |
| Dupont                  | 50     | 19     | 31     |                      |
| United Catalysts        | 62     | 21     | 17     |                      |
| Haldor Topsoe<br>MK-121 | >55    | 21-25  | 8-10   |                      |



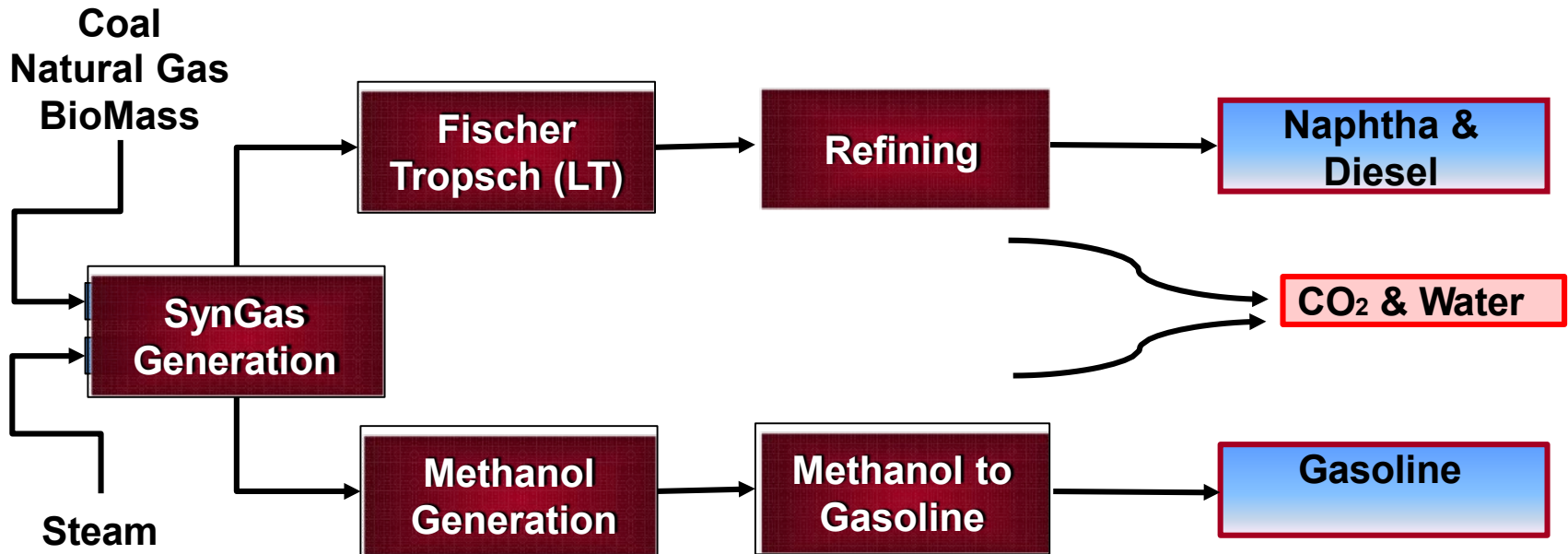


# Gas-to-Liquid (GTL) Technologies

**MTG Process**

**Fischer-Tropsch Synthesis**

# Routes for Synthetic Transportation Fuel



- Both are 3 Step Process
- Thermal efficiencies are essentially governed by initial C/H ratios... **Fuels are  $\sim\text{C}/\text{H}_2$** 
  - Coal is CH to Fuels plus CO<sub>2</sub>
  - Natural gas is CH<sub>4</sub> to Fuels plus H<sub>2</sub>O

Both the MTG and Fisher-Tropsch processes convert coal/NG into synthesis gas as an intermediary before producing the final products. However, their respective product slates are very different.

The Fisher-Tropsch process produces a broad spectrum of straight-chain paraffinic hydrocarbon which requires upgrading to produce finished products such as gasoline, jet fuel, diesel fuel, and lube base stocks. Due to the complexity of the product distribution, the economic justification for further upgrading/ processing of all the products improves for large scale projects (e.g. 50-80K BPD.)

MTG, in contrast, selectively converts methanol to high quality gasoline with virtually no sulfur and low benzene which can be sold as it is or blended in the refinery gasoline pool. About 90% of the hydrocarbon in methanol is converted to gasoline as the single liquid product, with the remainder primarily LPG.

# **Mobil Technology (Mobil's MTG Process)**

**Fixed/Fluidized bed reactor,  
360-415°C and 22 bar,  
Catalyst: ZSM-5**

Reference:

***Methanol to Gasoline: Production of Clean Gasoline from Coal***

***ExxonMobil Research and Engineering***

# Methanol to Gasoline Chemistry



**Overall exothermic reaction,  $\Delta H = - 55.68 \text{ kJ/mol}$**

# MTG Yields and Properties/Composition

**MTG GASOLINE YIELDS**

|                  | Percent of Feed | Percent of Hydrocarbon Product |
|------------------|-----------------|--------------------------------|
| Gas              | 1%              | 2%                             |
| LPG              | 5%              | 11%                            |
| Gasoline         | 38%             | 87%                            |
| H <sub>2</sub> O | 56%             | -                              |

**MTG GASOLINE  
PROPERTIES/COMPOSITION**

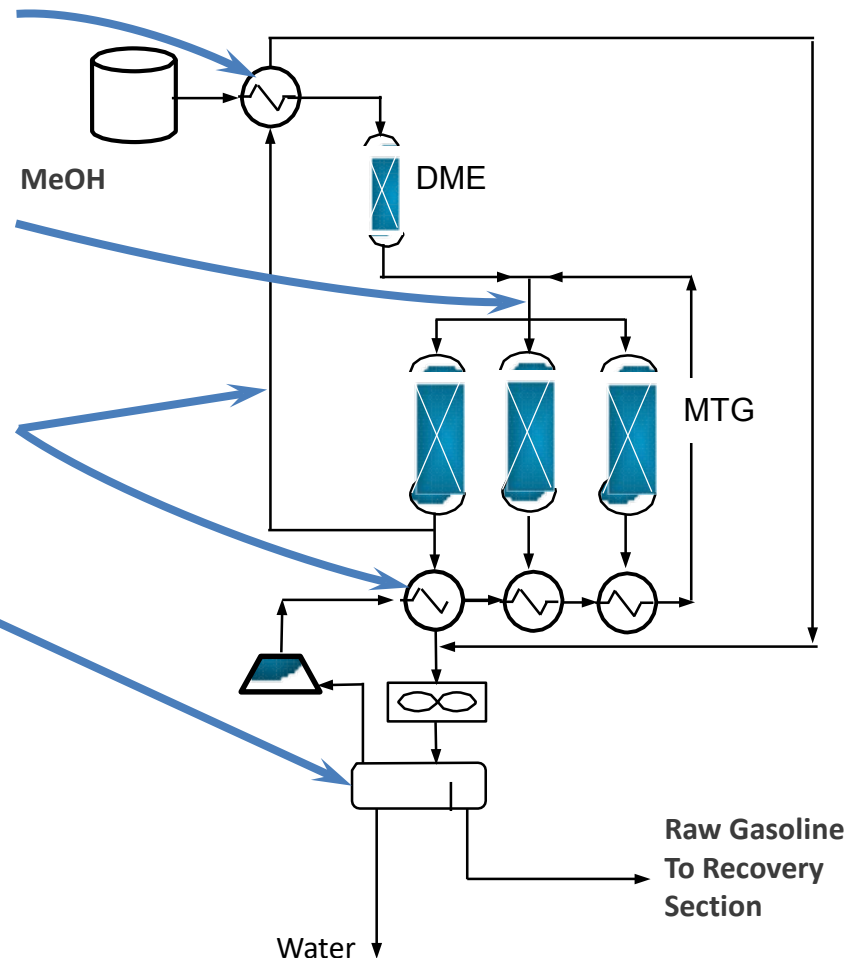
|                  |     |
|------------------|-----|
| Octane, RON      | 92  |
| Octane, MON      | 82  |
| (R+M)/2          | 87  |
| Paraffins, vol%  | 53  |
| Olefins, vol%    | 12  |
| Naphthenes, vol% | 9   |
| Aromatics, vol%  | 26  |
| Benzene, vol%    | 0.3 |
| Sulfur           | nil |

# ExxonMobil MTG Process:

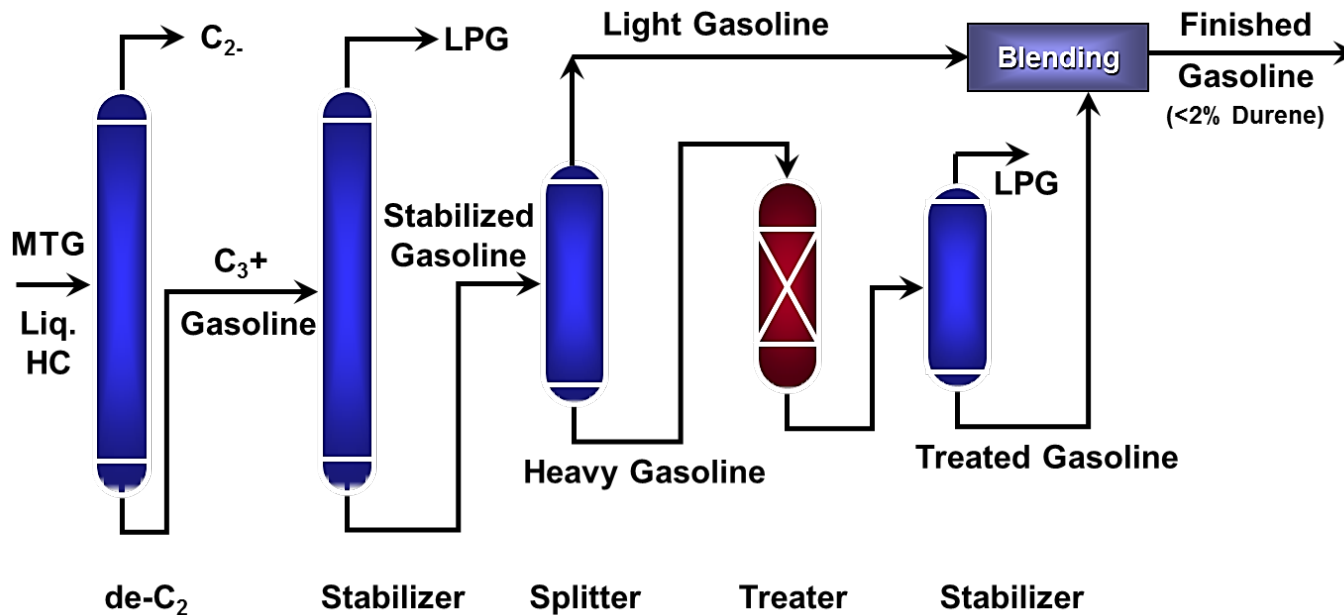
## MTG Reactor Section

- Methanol is vaporized by heat from the MTG reaction and fed into a DME reactor
- DME reactor effluent is mixed with recycle gas & fed to the MTG reactors
- Recycle gas controls reactor temperatures
- MTG reactor effluent is cooled by the methanol feed, recycle gas & air or water
- Condensed effluent is separated into gasoline, water and gas
- Raw gasoline is pumped to a recovery section

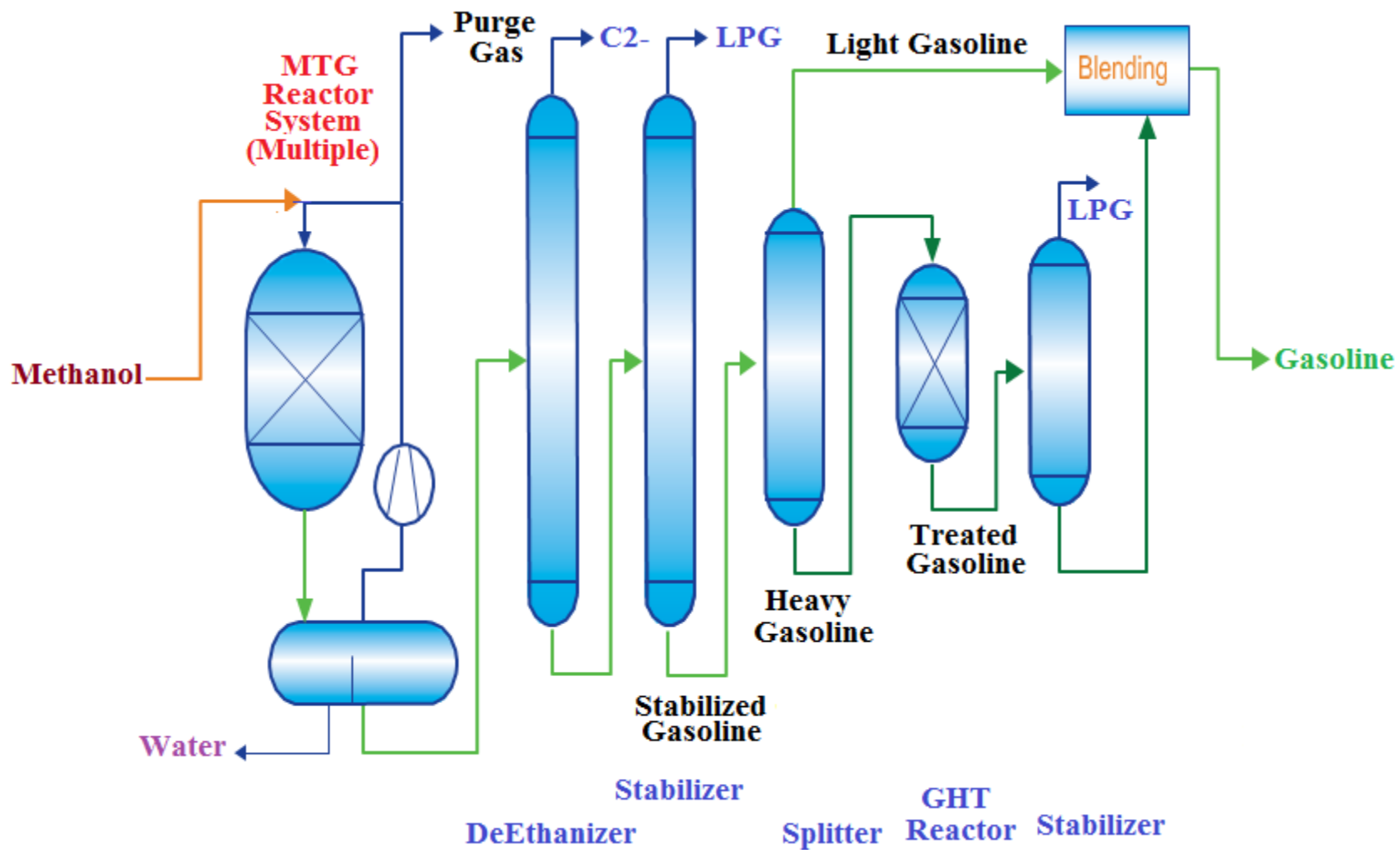
Basic MTG Process Flow



# Raw Gasoline Recovery Section



- Raw MTG gasoline is fed to de-ethanizer and stabilizer columns to remove fuel gas and LPG fractions from the gasoline
- Stabilized gasoline is split into light and heavy gasoline fractions
- Heavy gasoline is mildly hydro-treated to reduce Durene (1,2,4,5 tetra-methyl benzene) content (to 2% maximum)
- Treated heavy gasoline and light gasoline are blended into finished product



# Methanol to Olefins (MTO)

## Mobil Technology (Mobil's MTO Process)

**Fixed/Fluidized bed reactor,**

**470-515°C and 3.0-4.5 bar**

**Catalyst: ZSM-5**

## **UOP Process**

**Fluidized bed reactor**

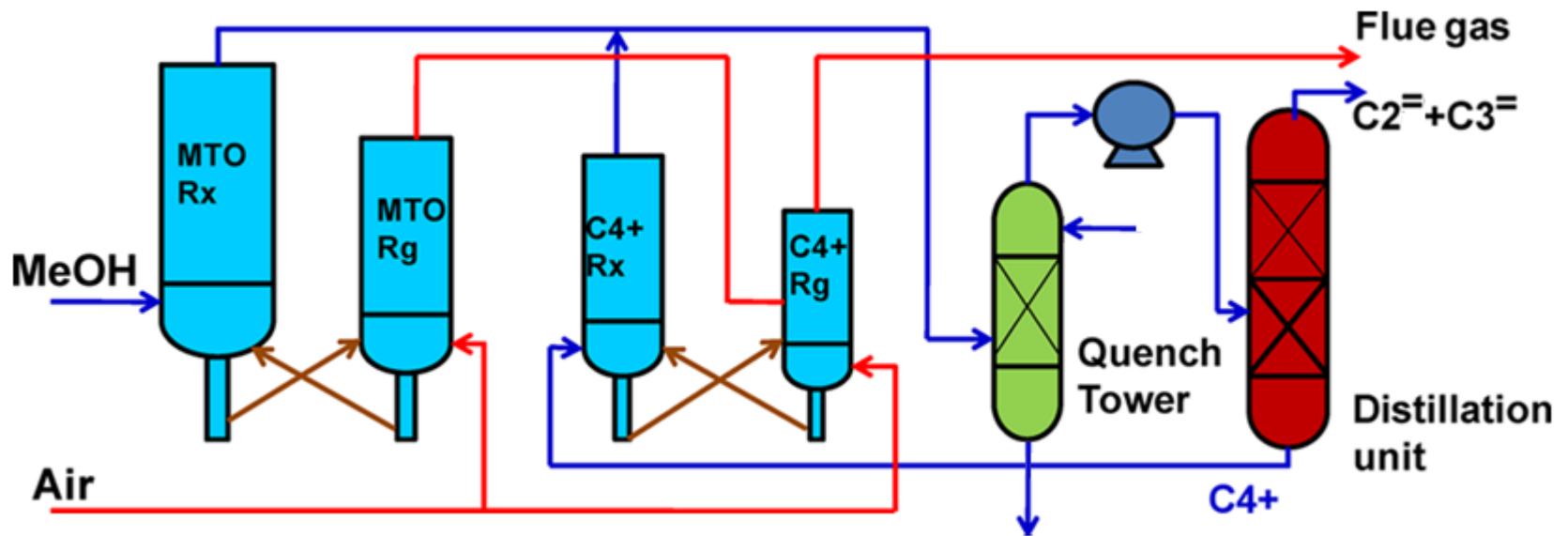
**Catalyst: SAPO-34**

Methanol to Olefins (MTO): From Fundamentals to Commercialization.  
*ACS Catalysis*, **2015**, 5(3), 1922-1938.

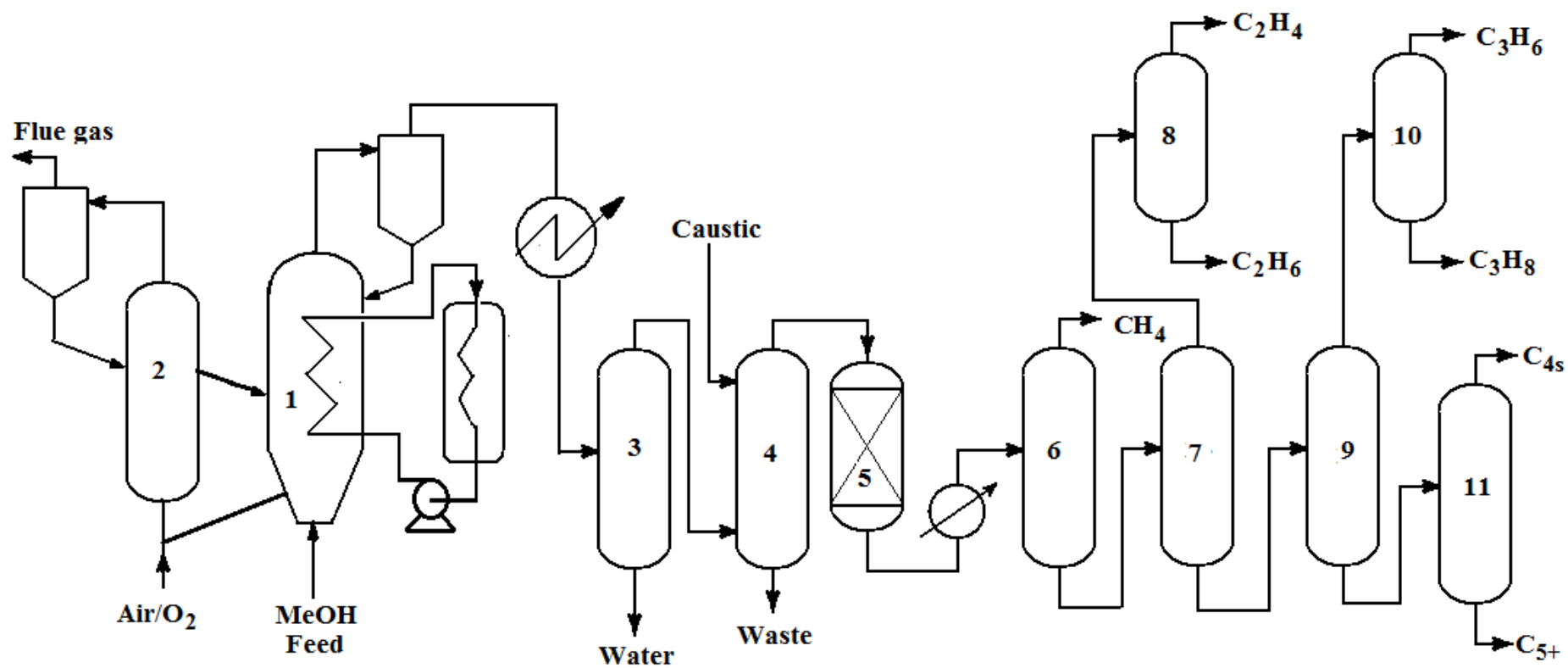
- High selectivity and yield to light olefins.

|                                        |                  |
|----------------------------------------|------------------|
| <b>Material</b>                        | <b>SAPO-34</b>   |
| T atom %                               | 10% Si (gel)     |
| <b>Selectivities (2 hr)</b>            |                  |
| C <sub>2</sub> -C <sub>4</sub> olefins | 96               |
| CH <sub>4</sub>                        | 1.4              |
| C <sub>2</sub> H <sub>6</sub>          | 0.3              |
| C <sub>3</sub> H <sub>8</sub>          | 0.9              |
| <b>Stability</b>                       |                  |
| hr at >50% conversion                  | >40              |
| <b>Coking</b>                          |                  |
| carbon on used catalyst                | 19% after 54 HOS |

ZSM-5 → Has sodium, if u want acidic catalyst replace Na by H & get HZSM-5 catalyst



## MTO Process Flowsheet



1: Reactor; 2: Regenerator; 3: Separator; 4: Caustic washer; 5: Drier; 6: Demethanizer; 7: Deethanizer;  
8: C<sub>2</sub>-Splitter; 9: Depropanizer; 10: C<sub>3</sub>-Splitter; 11: Debutanizer

# Methanol to Propene (MTP)

## Lurgi's MTP Process:

**Lurgi MTP™ - Methanol-to-Propylene (MTP) is an on-purpose technology for producing propylene from methanol, independently of the original feedstock. The technology offers clients low cash costs, and its propylene product can be used as feedstock for a variety of petrochemical processes.**

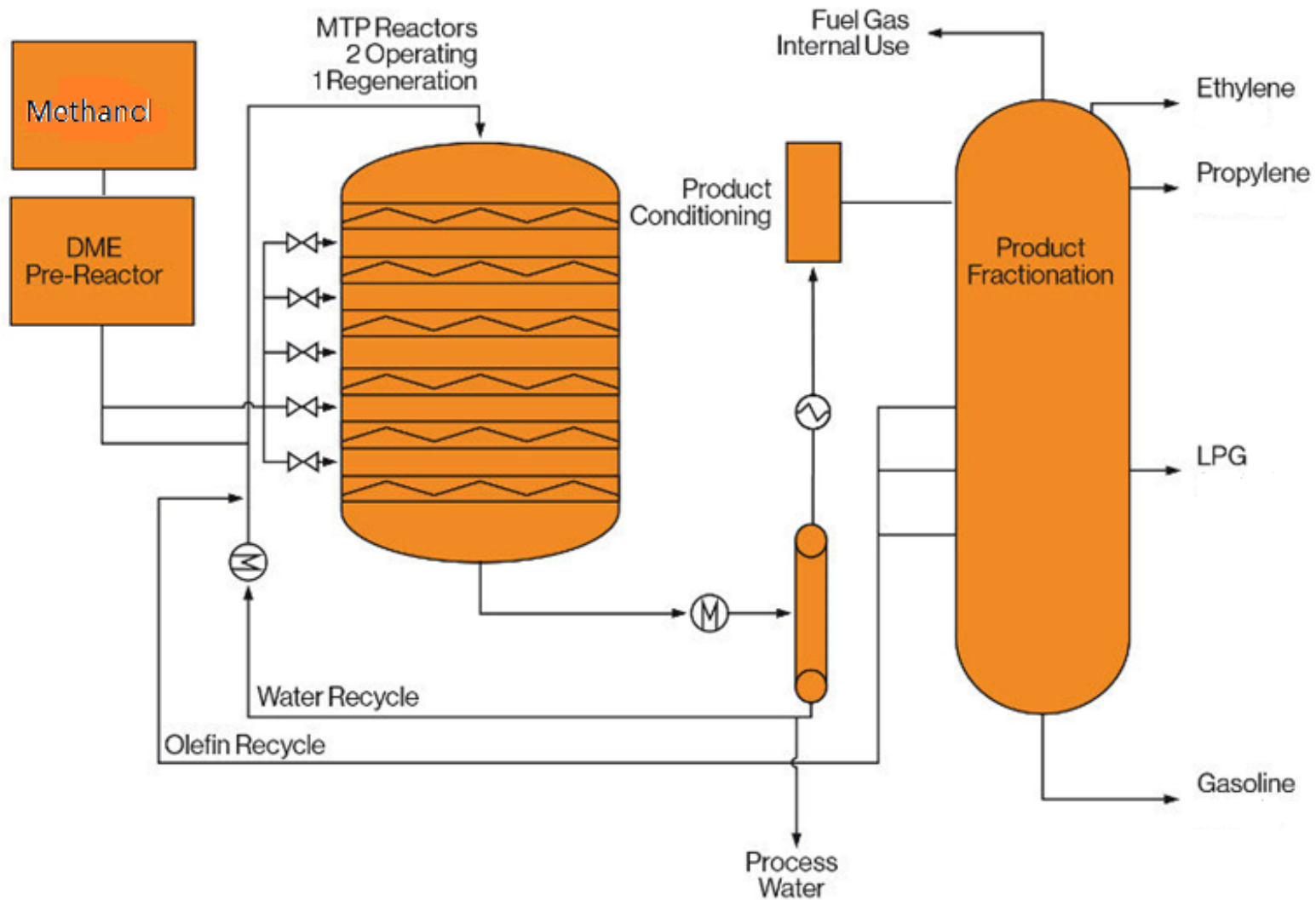
**ZSM-5 based catalyst developed by Sud-Chemie AG**

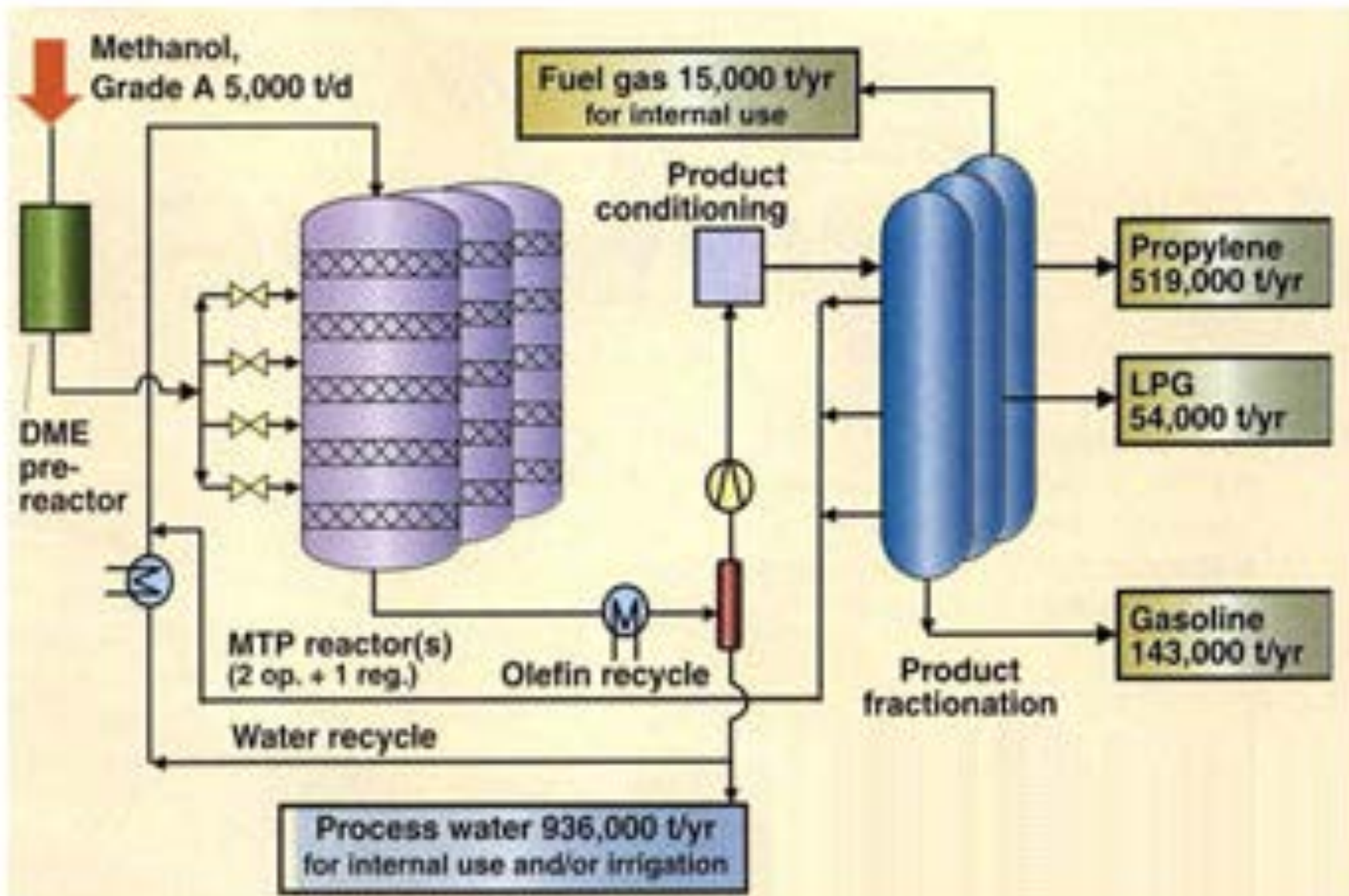
(Slower deactivation of this catalyst allows the use of a fixed bed reactor technology)

**Operates at 425 °C and 1.5 bar pressure**

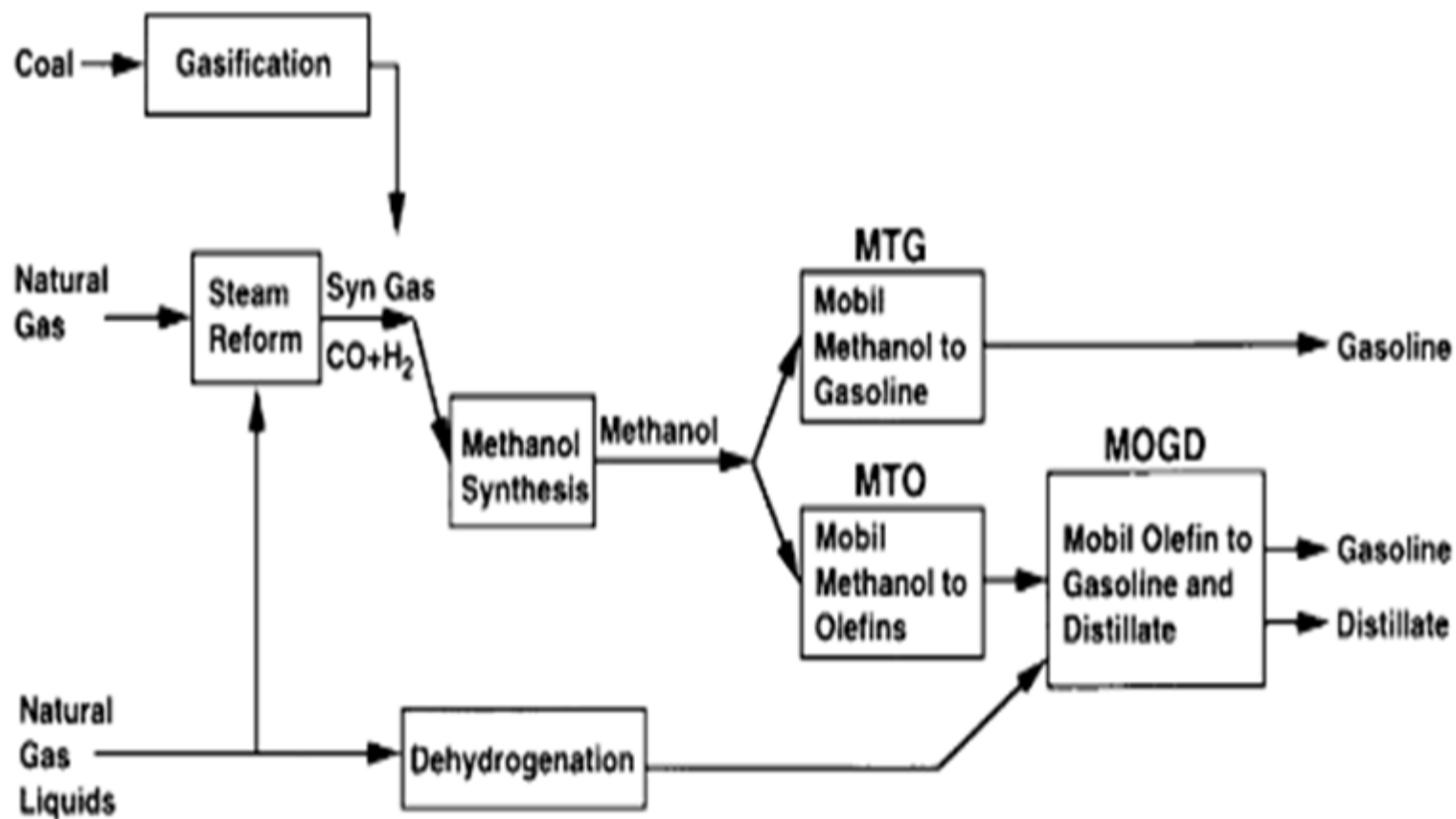
**Propene as the main product**

(70% yield and 97% selectivity)





# Mobil's Olefin-to-Gasoline and Distillate (MOGD) Process



**Gasoline and Distillate Production via Methanol and Mobil's ZSM-5 Technology**

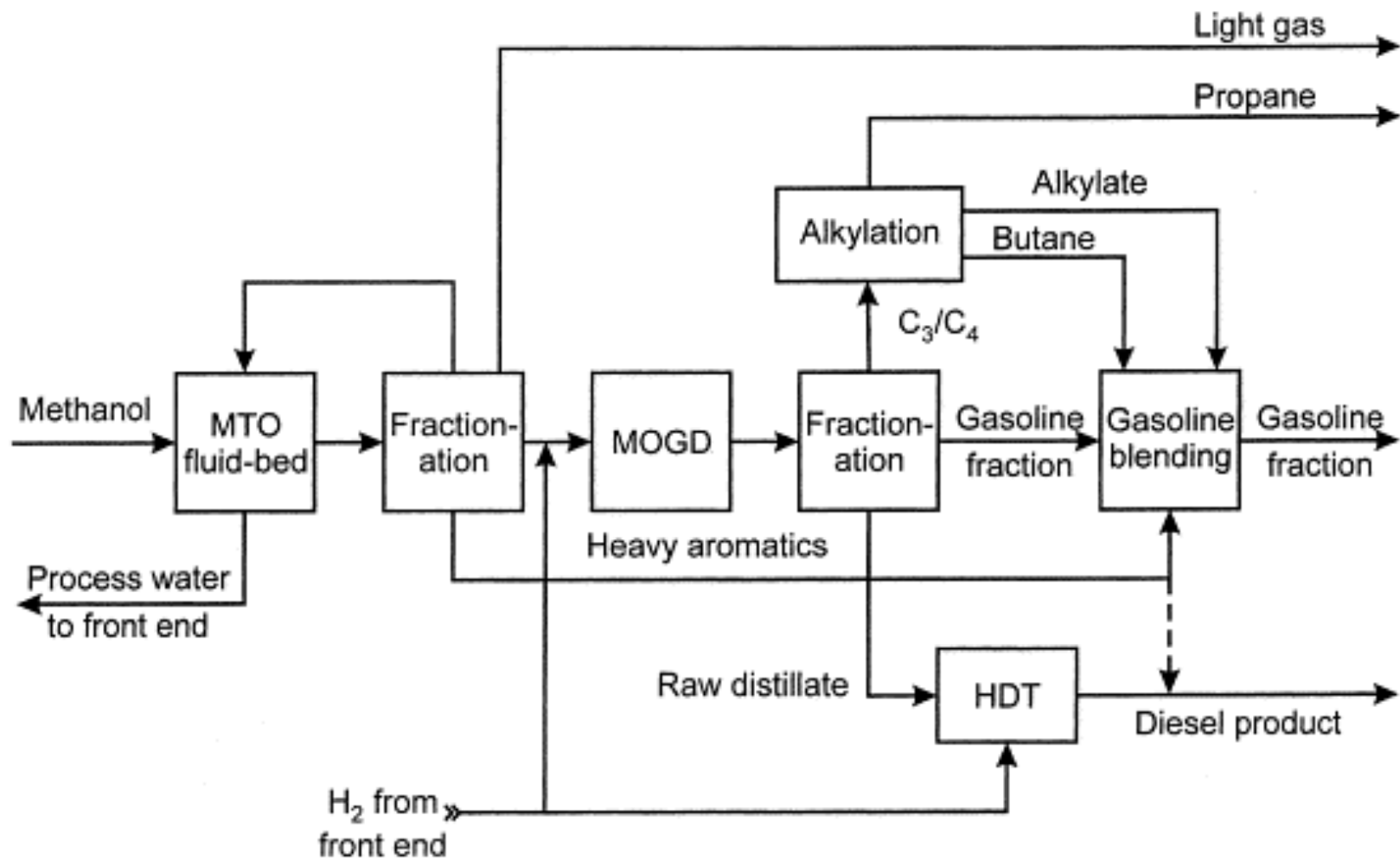
**Both the processes use ZSM-5 as catalyst and the combined MOGD process offers a high valuable gasoline (C5 to C10) and Distillate/diesel (C10 to C20) yield in various proportions – starting from light olefins (C3 to C4).**

**The gasoline and distillate/diesel selectivity is larger than 95% of the olefins in the feed and the gasoline/distillate product ratios range from 0.2 to >100.**

**The distillate product is mostly isoparaffinic and is an exceptionally good blending stock due to its high cetane index, low pour point and negligible sulphur content.**

**Conditions: 40 to 330 °C / 0.07 to 140 bar**

**Primary effect of the process variables is to alter the average product molecular weight.**

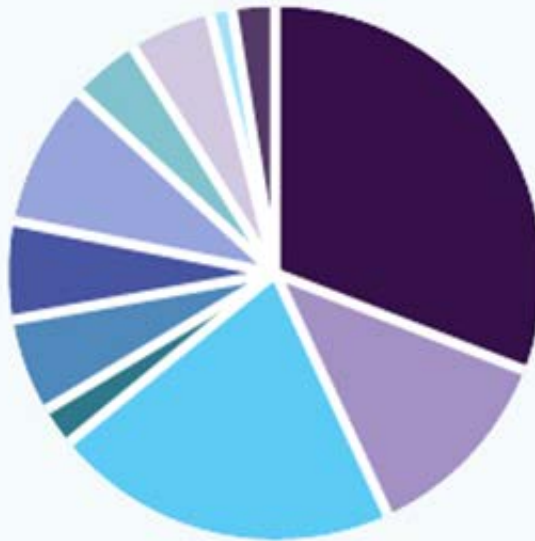


# Formaldehyde from Methanol

Formaldehyde is one of the most important chemical products worldwide (production of  $\sim 4\text{-}5 \times 10^7$  tons per year) for the production of urea-phenolic, acetal and melamine resins.

## Global Formaldehyde Market

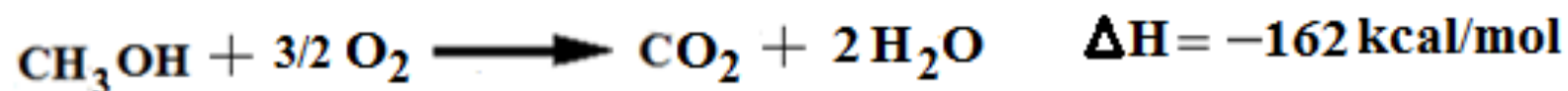
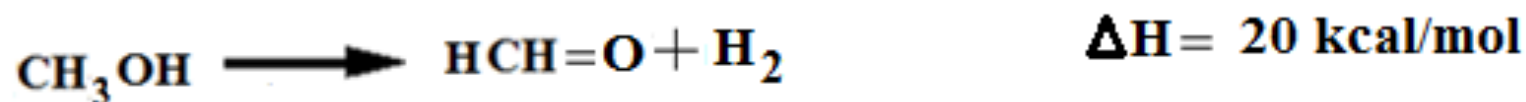
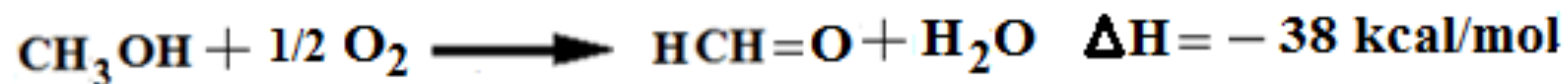
share, by end-use, 2020 (%)



**\$7.8B**

Global Market Size,  
2020

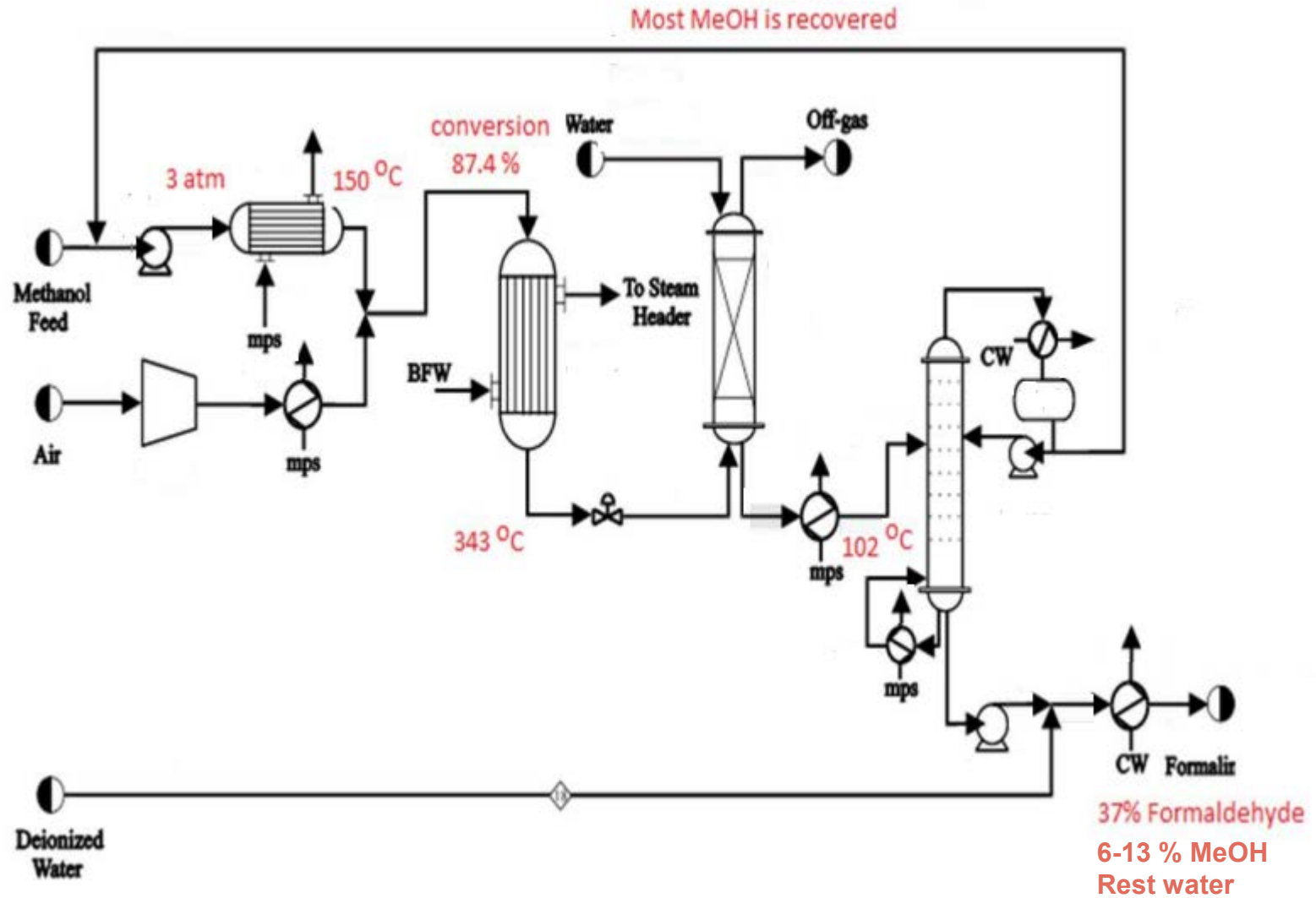




**Catalyst:** Silver gauze or Finely made Copper gauze

**450 – 600 °C, Atmospheric pressure**

**MeOH:Air ratio = 0.3 – 0.5**



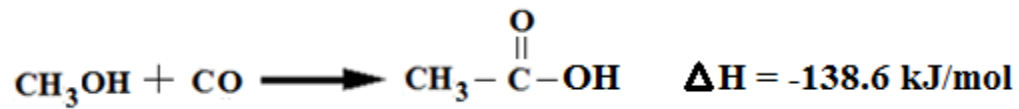
## **Acetic Acid from Methanol**

**In 2021, market volume of acetic acid worldwide amounted to over 16.7 million metric tons. It is forecast that the market volume of this organic compound will grow to around 22.6 million metric tons worldwide in the year 2029.**

**Acetic acid is used to prepare vinyl acetate, cellulose acetate, and a miscellany of commercial acetate esters including butyl, ethyl, n-propyl, and iso-propyl. However, the largest volume esters are acetates of glycol ethers. Acetic acid is also used as a solvent in the Amoco process for terephthalic acid.**

**The preferred industrial method for its manufacture is by the carbonylation of methanol and this accounts for approximately 60 per cent of the total world acetic acid manufacturing capacity.**

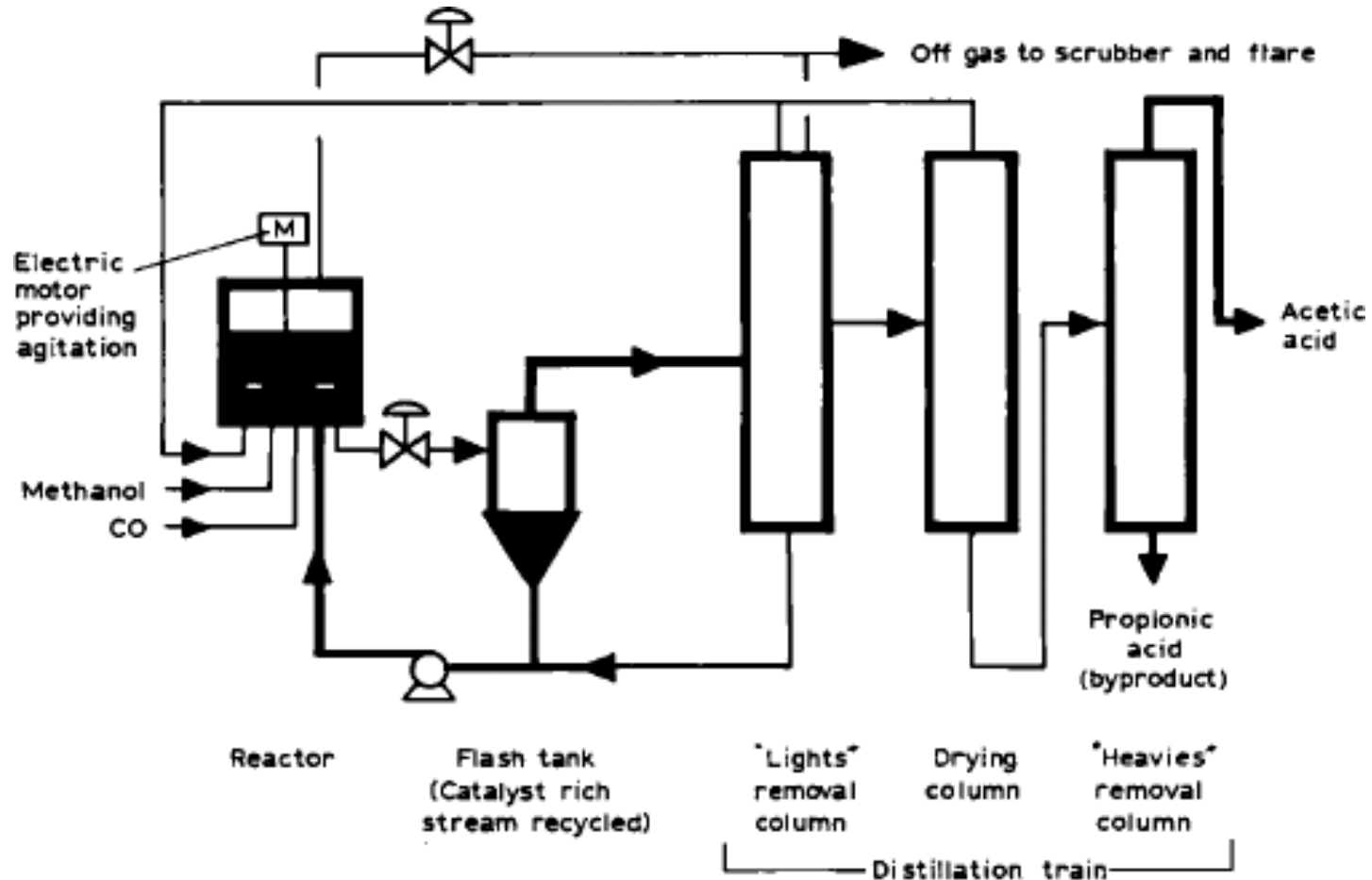
**The carbonylation of methanol, catalyzed by rhodium, was invented by Monsanto in the 1960s and for 25 years was the leading technology. In 1996 a new, more efficient, process for the carbonylation of methanol was announced by BP Chemicals, using an iridium catalyst.**



**Catalyst: Iodine promoted Rhodium**

**200 °C, 30-60 atm**

**99%+ selectivity for acetic acid based on methanol**

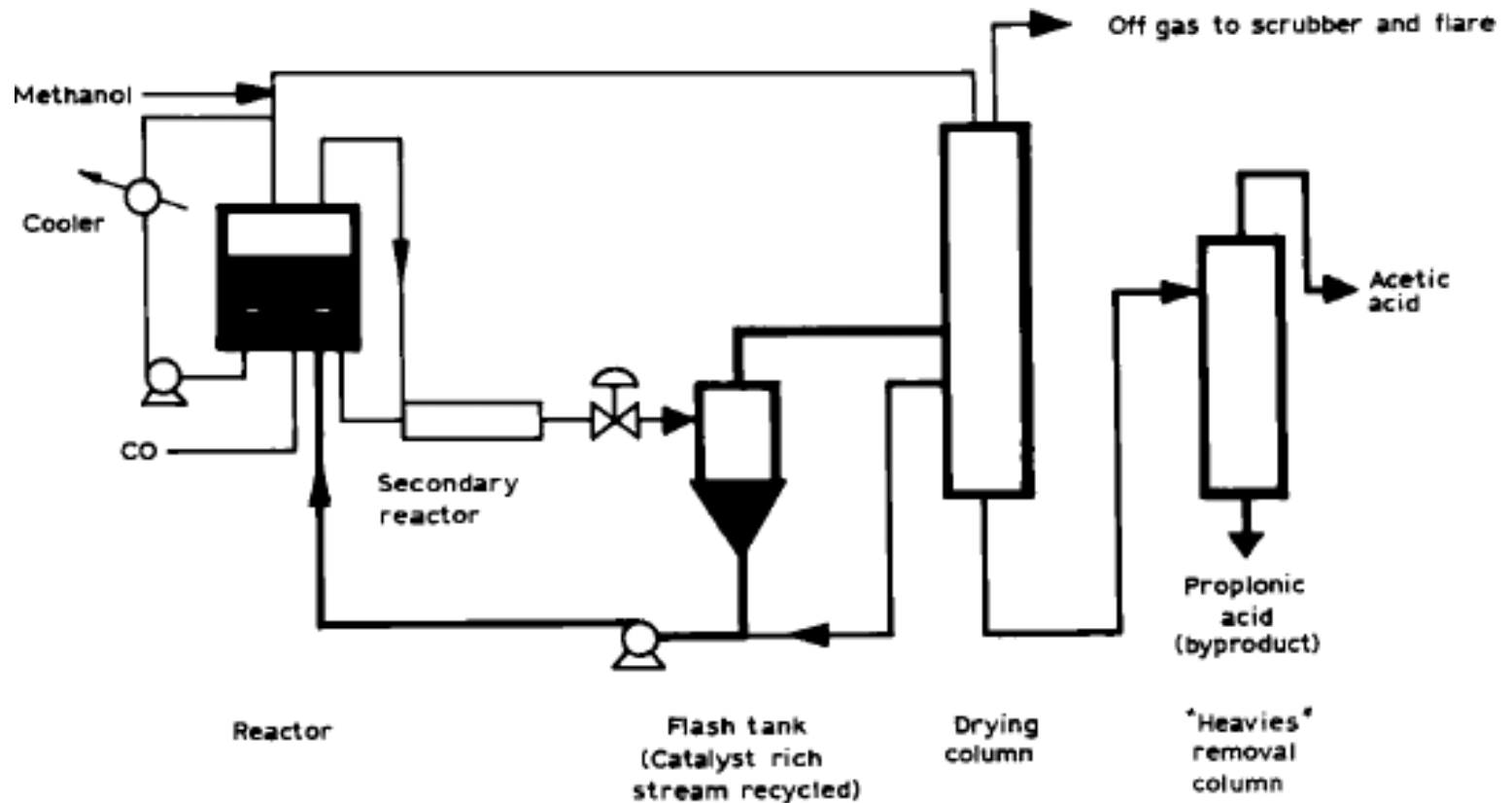


## Cativa™ Process

Very similar to the Monsanto Process

Iridium-based catalyst

Advantages are higher stability of the catalyst, less liquid by-products.

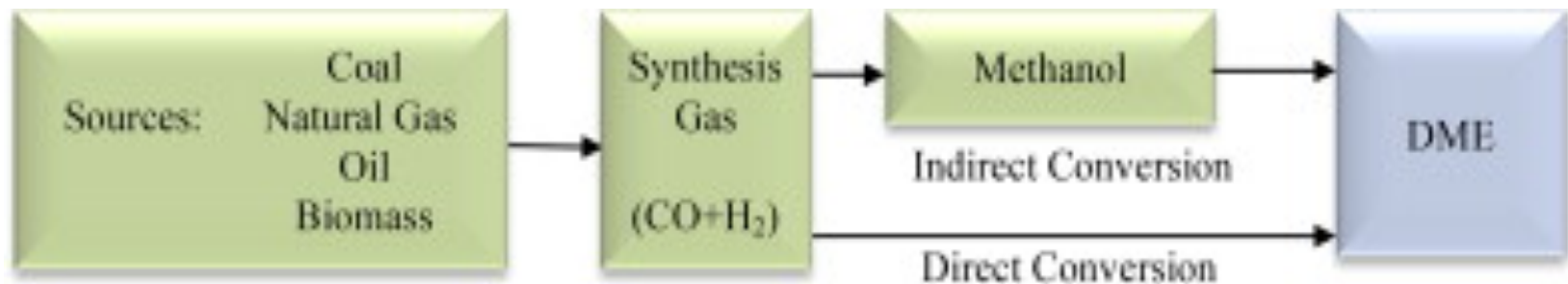


## Dimethyl Ether (DME)

DME is used primarily as a propellant.

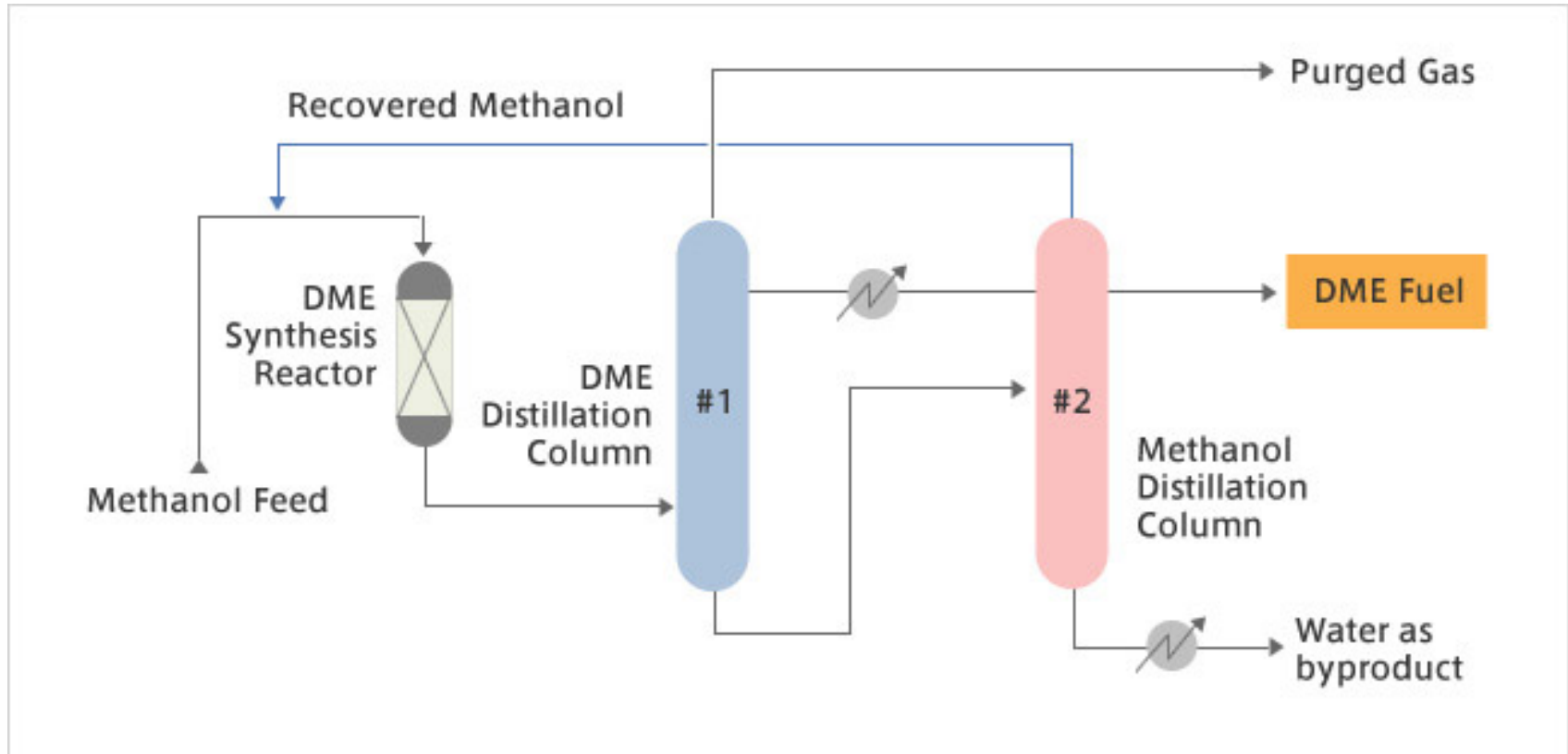
DME is miscible with most organic solvents and it has a high solubility in water.

The use of DME as a fuel additive for diesel engines has been investigated due to its high volatility (desired for cold starting) and high cetane number.





$$\Delta H = -23.5 \text{ kJ/mol}$$



## Production of Dimethyl Ether from Methanol

## Syngas to DME (STD)

**Methanol synthesis from CO:**



**Methanol dehydration:**



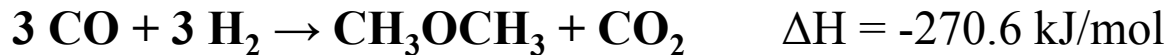
**Water gas shift reaction:**



**Methanol synthesis from CO<sub>2</sub>:**



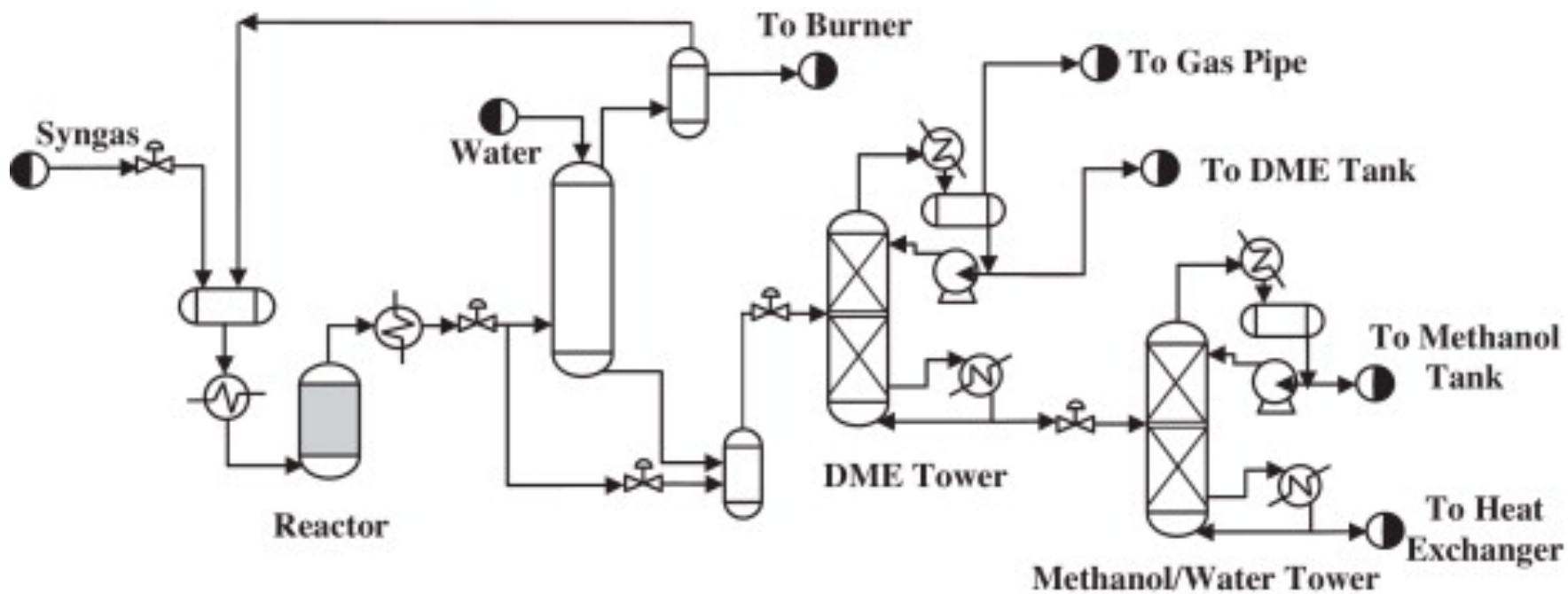
**Overall reaction:**



**Haldor-Topsoe Process:**

Bifunctional catalyst: Addition of HZSM-5 to the traditional CuO/ZnO/Al<sub>2</sub>O<sub>3</sub> methanol synthesis catalyst

290 – 400 °C; 80-120 bar



## Direct synthesis of DME from syngas (STD)

Thank you